

Harnessing Disagreement to Create AI-Powered Systems That Reflect Our Values

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ABSTRACT

How do we build artificial intelligence systems that reflect our values? Competing potential values we may want to choose from are, at their core, made up of disagreements between individual people. But while the raw datasets that most ML systems rely on are made up of individuals, today's approaches to building AI typically abstract the individuals out of the pipeline. My thesis contributes a set of algorithms and interactive systems that re-imagine the pipeline for designing and evaluating AI systems, requiring that we deal with competing values in an informed and intentional way. I start from the insight that at each stage of the pipeline, we need to treat individuals as the key unit of operation, rather than the abstractions or aggregated pseudo-humans in use by today's approaches. I instantiate this insight to address two problems that arise from today's AI pipeline. First, evaluation metrics produce actively misleading scores about the extent to which some people's values are being reflected. I introduce a mathematical transformation that more closely aligns metrics with the values and methods of user-facing performance measures. Second, the resulting AI systems either surreptitiously choose which values to listen to without input from users, or simply present several different outcomes to users without mechanisms to help them select an outcome grounded in their values. I propose a new interaction paradigm for deploying classifiers, asking users to compose a jury consisting of the individual people they'd like their classifiers' decisions to come from.

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1 INTRODUCTION

Competing potential values are central to our most important and compelling problems, ranging from disagreement within an online community about what sort of speech should be allowed [15], to disagreement among doctors about how to form a healthcare diagnosis [24], to disagreement among musicians about how to interpret a symphony [26]. These competing values are, at their core, made up of disagreements between individual people. But today's

approaches to building AI typically abstract individual people out of the pipeline, resulting in two fundamental problems. First, evaluation metrics produce actively misleading scores about the extent to which some people's values are or are not being reflected. Without effective metrics, making progress has become akin to hill-climbing on noise. Second, the resulting AI systems either surreptitiously choose which values to reflect without input from users – which is clearly problematic – or simply present several different outcomes to users without mechanisms to help users act on them in a way that's grounded in their values. As a result of these problems, while modern deep learning techniques now outperform human judgment on many AI tasks such as image classification, the AI-powered systems that aim to address many problems in social computing and HCI remain ineffective at best and highly problematic at worst.

My thesis contributes a set of algorithms and interactive systems that re-imagine the pipeline for designing and evaluating AI systems, requiring that we deal with competing values in an informed and intentional way. Specifically, we should be informed about the consequences that disagreement has on our systems' ability to reflect our values, and we should make intentional choices about which values our systems are or aren't reflecting. I start from the insight that at each stage of the pipeline, we need to treat individual people as the key unit of operation, rather than the abstractions or aggregated pseudo-humans in use by today's approaches, which can give rise to the ecological fallacy [9].

My thesis develops these ideas through three projects: two centered around evaluation, and a third around deployment. The first, the disagreement deconvolution, shows how existing ML classification metrics produce scores that are substantially and falsely inflated above actual user-facing performance because they only reflect the values of a subset of the population, and then introduces a mathematical transformation that more closely aligns metrics with the values and methods of user-facing performance measures [10]. The second, HYPE (Human eYe Perceptual Evaluation) demonstrates that by grounding annotation task designs in psychophysics methods from perceptual psychology, we can enable a meaningful evaluation metric for models attempting a nuanced, high-dimensional task for which reasonable people disagree [33]. The third, Juries, asks how we might rearchitect AI systems for situations where irreconcilable disagreements between groups make it difficult to define an unambiguous 'ground truth' label. It proposes a new interaction paradigm for deploying classifiers, asking users to compose a jury consisting of the individual people they'd like their classifiers' decisions to come from. In this abstract, I summarize each of these three projects.

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2 THE DISAGREEMENT DECONVOLUTION

The machine learning classifiers that underpin modern social computing systems such as Facebook, Wikipedia, and Twitter are a study in contrasts. On one hand, current performance metrics for popular tasks such as comment toxicity classification and disinformation detection are extremely high, featuring up to .95 ROC AUC [3, 34], making the problem appear nearly solved. On the other hand, audits suggest that these algorithms' performance is in reality much poorer than advertised [21]. Even well-intentioned platforms will continue to make highly-publicized mistakes [6] until they can better align metrics with reality.

This disconnect between classifier performance and user-facing performance is indicative of a larger disconnect between how machine learning (ML) and human-computer interaction (HCI) researchers evaluate their work. ML aims to evaluate technical performance, developing metrics that measure generalization error over unseen examples such as precision, recall, and ROC AUC. HCI instead aims to report user-facing experience, developing metrics that measure direct user response or opinion such as agreement rates, Likert scales, and behavioral outcomes. In many cases, common metrics used for technical performance are already aligned with user-facing performance, as strong gesture recognition [31] or text entry prediction rates [29] also represent strong user feedback. However, in the common ML metrics used for several critical tasks in social computing and HCI scenarios such as comment toxicity, the labels used to evaluate generalization error often do not directly map to user responses or opinions. As a result, strong technical performance over common metrics appears substantially and falsely inflated above user-facing performance.

In this project, we introduce a transformation that more closely aligns ML classification metrics with the values and methods of user-facing performance measures [10]. We base our transformation on the observation that common technical performance metrics flatten multiple opinions into a single ground truth label for each example, incorrectly ignoring disagreement between people [17, 27]. For an HCI researcher or professional who is measuring perceptions of toxicity, the quantity of interest is rarely a single bit of *whether* a given comment is toxic. At the very least, they'd like to know *the proportion of people* who view the comment as toxic. The ML practice of aggregating multiple annotators' responses for each example via majority vote or other approaches [25] into a single ground truth label [1] makes sense when the whole population tends to agree on the answer to the question, such as "Does this image contain a cat?" in object classification or "Does the word 'java' in this sentence refer to coffee or the programming language?" in word sense disambiguation. However, social computing tasks and other common tasks in HCI often remain deeply contested, with much lower levels of agreement amongst the population [4]. Our goal is to incorporate an estimate of how contested each label will be into ML metrics in an interpretable way, producing metrics that ask not "What proportion of ground truth labels does the classifier agree with?" but "What proportion of the population does the classifier agree with?" For traditional ML tasks such as object classification, the resulting metrics would likely be very similar; but in social computing tasks and many others in HCI, the measurements could diverge substantially.

We establish that disagreement places fundamental limits on applying today's widely adopted supervised learning pipeline to these problems. We find that social computing tasks cannot possibly achieve performance remotely comparable to that of classic tasks like image detection. For classic tasks, the disagreement-adjusted scores remain near perfect, but for social computing tasks, these scores drop dramatically. For instance, in the popular Jigsaw toxicity task where Jigsaw claims that models achieve 0.95 ROC AUC, we find that even an oracle classifier that perfectly predicts each aggregated annotation could only achieve a disagreement-adjusted ROC AUC of 0.73. We show how these fundamental limits are inherently hidden by following the very pipeline that causes them.

3 HYPE: A BENCHMARK FOR HUMAN EYE PERCEPTUAL EVALUATION OF GENERATIVE MODELS

The disagreement deconvolution aimed to transform existing machine learning metrics such that they communicate how a model can perform in the presence of disagreement, in part by designing an algorithm to help disentangle annotator beliefs from noise. However, for such an algorithm to result in useful metrics, the beliefs themselves must be worth disentangling. The disagreement deconvolution left this entirely to existing work, relying on existing datasets with task designs created by others. With HYPE, we reverse that focus. We ask: given a high-dimensional, nuanced task in which many annotators reasonably can and should disagree, how can we design an annotation task such that we learn something meaningful about each annotators' beliefs?

Specifically, we targeted a focal task for measuring the progress of generative models: generating realistic images of fake faces [33]. Automated metrics widely in use today are either heuristic approximations [5, 8, 12, 20, 22, 23] or intractable density estimations, examined to be inaccurate on high dimensional problems [2, 11, 28]. Realizing the constraints of available automated metrics, many generative modeling tasks resort to human evaluation and visual inspection [8, 22, 23]. However, human evaluations, such as those given on Amazon Mechanical Turk [8, 22], remained effectively meaningless because they failed to separate models from one another. In effect, researchers were asking annotators the wrong questions, basing their task design on existing approaches that had been effective for classic image labeling tasks with a clear correct answer. Instead, we require a task design that can meaningfully and reliably capture each annotator's judgements for a high-dimensional task where there is no correct answer.

We proposed to measure the perceptual realism of generative model outputs via a procedure inspired by psychophysics methods in perceptual psychology literature [7, 13]. We display images using adaptive time constraints to determine the time-limited perceptual threshold a person needs to distinguish real from fake. The resulting score is understood as the minimum time, in milliseconds, that a person needs to see the model's output before they can distinguish it as real or fake. Future work will extend HYPE to additional generative tasks such as text, music, and video generation.

4 JURIES

We’ve now presented two projects focused on evaluation in the presence of disagreement, both of which work towards our goal of requiring that we are informed about the consequences that disagreement has on our systems’ ability to reflect our values. With our final project, Juries, we shift our focus away from evaluation and towards deployment, aiming to require that we are not only informed about the consensus of competing values, but make *intentional*, explicit choices about which values our systems are and are not reflecting.

For instance: in social computing systems such as Facebook, Wikipedia, and Twitter as well as classification tasks ranging from content moderation to hate speech detection to misinformation detection [3, 34], there often exists fundamental disagreements between majority and minority groups on what the correct labels ought to be. For example, women and Black users of Twitter, who are more commonly targets of online harassment, observe harassment that White males might not view as problematic. The resulting AIs appear to have excellent performance on held-out test sets [10], and then launch to the public—where they fail marginalized groups. Advances in algorithmic fairness can help reduce disparate outcomes [16], there is little guidance on *what* labels we should be imitating when there are irreconcilable disagreements among groups in society.

Today, when deploying machine learning models for tasks with disagreement, such as the scenario described above, the technical approaches taken by researchers and practitioners result in two possible outcomes. In the first, particularly common for many tasks in social computing, the model will surreptitiously “resolve” disagreement without input from users, because it has been trained to provide a single label. This is to say, the model predicts a single probabilistic score, which may or may not represent annotator disagreement (it may also represent the model’s uncertainty for a variety of other reasons) [17, 27]. In the second, more recent approach, a model is explicitly trained to predict a distribution of possible scores for each label, representing the proportion of annotators who’d select each label [14, 18, 19, 30, 32]. These are sometimes accompanied by an explanation for each label (e.g., why would someone have selected each possible answer) [24]. The second approach is at least an improvement over the first, as it explicitly acknowledges the existence of disagreement. However, merely explaining the existence of disagreement does little to help a user decide what to actually do about it: it partially meets our goal of dealing with disagreement in an informed way, but does little to meet our goal of requiring that we deal with disagreement in an intentional way.

We propose to push researchers and practitioners to make the value judgements of whose voices to listen to, and in what proportion, entirely explicit. They should explicitly specify the individuals whose opinions are to be used for final prediction, and in what proportion, akin to composing a jury of annotators, a *polis* whose perspectives should determine the outcome. To enable this interaction, we design a novel user interface coupled to a novel machine learning architecture that builds on recent advances in both deep-learning based recommender systems and in transformer-based natural language processing. We will develop visualizations that

allow nontechnical users to see how an AI jury reacts to each input, to understand the diversity of perspective within groups and across groups, and to follow how the models representing individual jury members are aggregated into a composite response. Our current primary application domain for Juries is content moderation, where today’s resulting AIs appear to have excellent performance on held-out test sets [10], and then launch to the public—where they fail marginalized groups. The higher-level ideas in this work are applicable to a wide variety of domains, such as medical decision making or creative tasks.

5 CONCLUSION

In this abstract, I presented my approach to designing a machine learning pipeline for tasks with competing values, and described three projects that take this approach. Together, they begin to show that by treating individual, real people as the key unit of operation with the machine learning pipeline, we can create AI that deals with disagreement within AI-powered systems in an informed and intentional way, paving the way towards creating AI that more effectively supports us in solving problems in social computing and HCI.

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